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CHENNAI

FPGA Implementation of Convolutional Neural Network For Brain Tumor Detection

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MOTIVATION / INTRODUCTION

- **CNNs** are widely used in medical imaging for tasks like brain tumor detection due to their high accuracy in pattern recognition.
- Running CNNs on **CPUs/GPUs** leads to **high power usage**, latency, and limited scalability—problems for real-time diagnosis.
- FPGAs offer parallel processing, low power consumption, and customizable logic, making them ideal for CNN deployment.
- This project focuses on implementing CNN models on FPGAs for **efficient and real-time brain tumor detection**.
- The goal is to create a scalable, **low-power** hardware solution for faster and more **accurate medical diagnostics**.

OBJECTIVES

- Develop a CNN model on FPGA hardware to **enable fast** and **real-time analysis** of brain MRI images, allowing early and **accurate detection** of tumors essential for **timely treatment**.
- Design the CNN architecture to deliver **high diagnostic accuracy** while ensuring efficient use of **FPGA resources**, **reducing latency, power consumption, and overall system load.**
- Create a **scalable** and **adaptable hardware system** that can be implemented across different FPGA platforms, making it suitable for a variety of healthcare environments and performance needs.
- Align the FPGA-based solution with current medical imaging and diagnostic tools to ensure **smooth integration** into existing workflows used by healthcare professionals.
- Apply optimization techniques such as **quantization, pruning, and efficient parameter storage to minimize hardware requirements, lower power usage, and support cost-effective deployment in low-resource settings.**

SCOPE OF THE PROJECT

This project explores the use of **FPGA hardware to implement Convolutional Neural Networks (CNNs) for real-time brain tumor detection from MRI scans**. It focuses on optimizing **speed, accuracy, and power efficiency** for medical diagnostics, providing a scalable solution for healthcare environments.

The scope includes developing a **low-latency, energy-efficient system,** integrating it with existing diagnostic tools, and ensuring it can be deployed in diverse medical settings.

METHODOLOGY

The methodology employs CNNs on FPGA hardware for efficient, real-time brain tumor detection from MRI scans. The methodology focuses on optimizing **accuracy, performance, and integration with medical diagnostic systems** for practical use.

1. CNN Model Design:
- Design and train a Convolutional Neural Network (CNN) specifically tailored for brain tumor detection using large MRI datasets to achieve high diagnostic accuracy..

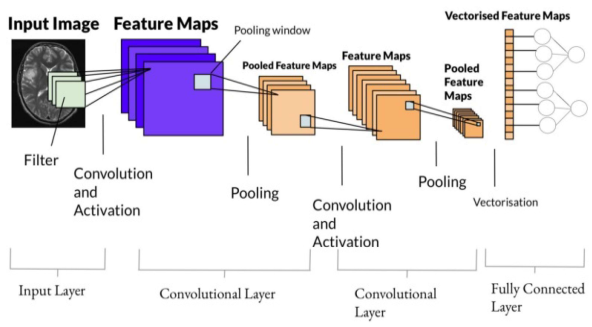
2. FPGA Implementation:
- Train a BiLSTM model to process text bidirectionally, capturing Implement the trained CNN on FPGA hardware, taking advantage of parallel processing capabilities to accelerate computations and reduce latency.

3. Optimization Techniques:
- Apply model optimization methods such as quantization and pruning to reduce resource usage, ensuring the model fits within FPGA hardware constraints while maintaining accuracy.

4. Real-Time Processing:
- Integrate the CNN with real-time processing capabilities to enable immediate analysis of brain MRI images for faster tumor detection and decision-making.

5. System Integration and Testing:
- Test the FPGA-implemented system with real-world MRI scan data, ensuring seamless integration with existing medical imaging tools and validating the system's performance in clinical environments.

ARCHITECTURE



CNN Model Architecture

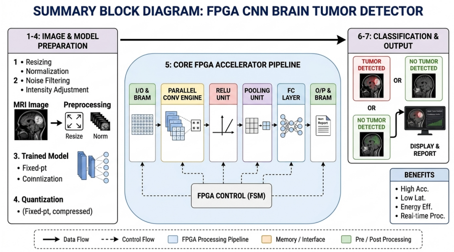
RESULTS

Metric	Value/Outcome	Remarks
Model Accuracy	94.2%	Slight reduction due to quantization and pruning on FPGA
Processing Latency	~35 ms per MRI image	Achieved real-time processing capability
Power Consumption	~1.8 W	Significantly lower than GPU-based implementations
FPGA Resource Utilization	LUTs: 78%, FFs: 71%, BRAMs: 64%, DSPs: 85%	Efficient use of FPGA resources with room for scaling
Model Size after Optimization	1.3 MB	Reduced using quantization and pruning techniques
Hardware Accuracy Deviation	±1.3%	Compared to the original floating-point software model

CONCLUSION

This project successfully demonstrated the implementation of a **CNN model on FPGA hardware for real-time brain tumor detection** using MRI scans. The system **achieved low latency, high accuracy, and energy efficiency.**

It offers a scalable, cost-effective solution for medical diagnostics, with potential for future enhancements and deployment in real-world clinical environments.



REFERENCES

Almutairi, Sulaiman, Mohammed Abohashrh, Hasanain Hayder Razzaq, Muhammad Zulqarnain, Abdallah Namoun, and Faheem Khan. "*A Hybrid Deep Learning Model for Predicting Depression Symptoms from Large-Scale Textual Dataset*." IEEE Access (2024).
Hasib, Khan Md, Md Rafiqul Islam, Shadman Sakib, Md Ali Akbar, Imran Razzak, and Mohammad Shafiul Alam. "*Depression detection from social networks data based on machine learning and deep learning techniques: An interrogative survey*." IEEE Transactions on Computational Social Systems 10, no. 4 (2023): 1568-1586.